Northeastern University, Khoury College of Computer Science

CS 6120 — Some Extra Practice 6

Due: March 30, 2026(100 points)

Practice Quiz 6 - Advanced Sequence Modeling

Instructions: Closed book. No calculators. Show all work. All probabilities are given in standard (not log) space unless stated otherwise.

Practice Quiz: Masked Self-Attention (Decoder Logic)

Setup: Consider a sequence of three tokens ($N = 3$) with embeddings in $\mathbb{R}^3$:

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Assume identity projections for simplicity: $W_Q = W_K = W_V = I$. The scaling factor is $1/\sqrt{3}$.

Problem 1: The Score Matrix

Compute the raw dot-product score matrix $E = QK^\top$. *(Since $W_Q, W_K = I$, this is simply $XX^\top$).*

Problem 2: Causal Masking

In a decoder, we apply a mask M to E before the softmax to ensure token i cannot attend to token j if $j > i$.

$$M_{ij} = \begin{cases} 0 & \text{if } j \leq i \\ -\infty & \text{if } j > i \end{cases}$$

Write out the masked score matrix $\hat{E} = E + M$.

Problem 3: Scaled Softmax with Masking

Apply the scaling factor $1/\sqrt{3}$ to your result from Problem 2. Then, focus on the **second row** ($i = 2$). If the scaled, masked scores for the second row are $[\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\infty]$, compute the final attention weights for that row.

Problem 4: Conceptual Weighting

If a row in the attention weight matrix A is $[0.5, 0.5, 0]$, explain in one sentence what this means for the resulting output vector for that position.

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Quiz Solution Key

- **Q1:** $E = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$
- **Q2:** $\hat{E} = \begin{bmatrix} 2 & -\infty & -\infty \\ 0 & 1 & -\infty \\ 1 & 1 & 2 \end{bmatrix}$
- **Q3:** For row 2: The weights are $[\frac{e^{0/\sqrt{3}}}{e^{0/\sqrt{3}}+e^{1/\sqrt{3}}}, \frac{e^{1/\sqrt{3}}}{e^{0/\sqrt{3}}+e^{1/\sqrt{3}}}, 0]$.
- **Q4:** The output vector at that position will be an equal average of the first and second value vectors (v_1 and v_2), completely ignoring v_3.